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REVIEW

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The best solution is the simplest: advances in surgical and minimally invasive management of aneurysmal and simple bone cysts

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Abstract

Aneurysmal bone cysts (ABCs) and simple bone cysts (SBCs) are benign osseous lesions predominantly affecting children and adolescents, typically presenting with pain, deformity, or pathologic fracture. Despite their frequency, optimal management remains debated. This review synthesizes current evidence on therapeutic strategies, efficacy, recurrence, and safety. For ABCs, intralesional curettage remains the standard surgical approach, though recurrence rates vary and are reduced with adjuvants such as high-speed burring, cryotherapy, or cementation. En bloc resection offers the lowest recurrence; however, it is restricted by procedural morbidity and poor functional outcomes. Minimally invasive modalities, including sclerotherapy, selective arterial embolization, endoscopic curettage, and image-guided ablation, consistently achieve high rates of healing, pain relief, and functional recovery, often with fewer complications. Management of SBCs ranges from observation to intervention in symptomatic or unstable lesions. Elastic stable intramedullary nailing (ESIN) provides both decompression and structural stability, with high healing rates. Overall, advances in minimally invasive and biologic therapies are reshaping treatment paradigms, yet standardized outcome measures and head-to-head comparative trials remain needed.

Keywords Aneurysmal bone cysts, Simple bone cysts, Unicameral bone cysts, Intralesional curettage, Denosumab

Introduction

Aneurysmal bone cysts (ABCs) and simple bone cysts (SBCs) represent two of the most common benign osseous lesions encountered in orthopedic practice, particularly among pediatric and adolescent populations [1, 2]. Though both are non-cancerous, their clinical behavior, underlying pathophysiology, and management strategies differ substantially, posing distinct challenges for clinicians. ABCs are expansile, osteolytic lesions characterized by blood-filled cavities separated by fibrous septa, with a tendency for local aggressiveness and bone destruction [3]. In contrast, SBCs are developmental or reactive unicameral cavities filled with serous fluid [4]. Despite their benign nature, both lesions are clinically significant due to their association with pain, skeletal deformity, and a heightened risk of pathological fracture in young patients.

Management strategies for ABCs and SBCs have evolved substantially over the past two decades, driven by improved biologic understanding and the expansion

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of minimally invasive options [5–8]. In ABCs, recognition of USP6 rearrangements has refined diagnostic certainty and supported development of targeted approaches [9], while percutaneous modalities (e.g., sclerotherapy and SAE) have increasingly been adopted to reduce morbidity in anatomically complex lesions [10, 11]. Systemic therapy (most notably denosumab, a receptor activator of nuclear factor kappa-B ligand [RANKL] inhibitor) is now used selectively for aggressive, recurrent, or surgically inaccessible ABCs. However, outcome definitions and comparative evidence remain heterogeneous, limiting standardization of treatment selection and sequencing [12–15].

Given the persistent clinical challenges and the rapid evolution in the understanding and management of these lesions, a comprehensive review is warranted. The objectives of this review are (1) to summarize the distinct pathophysiology, molecular mechanisms, and diagnostic criteria of ABCs and SBCs; (2) to critically evaluate the spectrum of established and emerging treatment modalities, from minimally invasive techniques to open surgical procedures and targeted medical therapies; and (3) to highlight specific management challenges, particularly in pediatric populations and anatomically complex sites.

Epidemiology

ABCs have an estimated annual incidence of 0.14–0.32 per 100,000 individuals, with a slight female predominance and a median age of 10–13 years. Approximately 60% arise in long bones, most commonly in the metaphyseal regions of the femur, tibia, and humerus, although the spine and pelvis are also frequent sites [16–18].

SBCs, also known as unicameral bone cysts (UBCs), have a comparable annual prevalence of approximately 0.30 per 100,000 but have a stronger male predominance (male-to-female ratio up to 1.96:1) and typically present at a slightly younger median age of 9–11 years [17]. They most commonly involve the proximal humerus and femur and are frequently identified incidentally or after a pathologic fracture [19]. Both ABCs and SBCs are rare in the craniofacial skeleton and jaws; when present, they primarily affect young individuals and are usually asymptomatic [20, 21].

Diagnostic modalities of ABCs and SBCs

A precise diagnosis is the cornerstone of effective treatment for bone cysts. Understanding the distinct pathophysiology, molecular underpinnings, and characteristic imaging features of ABCs and SBCs is crucial for differentiating them from each other and, most importantly, from malignant mimics.

Histopathology and molecular pathogenesis

ABCs are blood-filled, multiloculated cystic spaces separated by fibrous septa. Histologically, these septa contain fibroblast-like stromal cells, numerous osteoclast-type giant cells, and reactive woven bone, often with osteoblastic rimming [18]. The cyst walls lack a true endothelial lining, though flattened endothelial-like cells may be present. The seminal discovery in ABC pathogenesis is the consistent presence of a *USP6* gene rearrangement in approximately 63–82% of primary cases [22–24]. Its most critical role is in distinguishing primary ABC from secondary "ABC-like changes" that can occur in a variety of other benign and malignant bone tumors, including giant cell tumor of bone, chondroblastoma, fibrous dysplasia, and, most importantly, telangiectatic osteosarcoma. These secondary changes mimic the histology of an ABC but lack the underlying *USP6* driver mutation [18]. In ABCs, *USP6* oncogenic activation occurs via promoter swapping, where a chromosomal translocation juxtaposes the entire coding sequence of *USP6* with the highly active promoter of a partner gene [25].

SBCs are fluid-filled cavities that are lined by a thin fibrous membrane composed of spindled to ovoid cells, with scattered osteoclast-like giant cells, foamy histiocytes, hemosiderin deposits, and cholesterol clefts. Fibrin-like collagen deposits and occasional woven bone may be present in the cyst wall [26]. Unlike ABCs, SBCs lack the prominent blood-filled spaces and septa with abundant giant cells. Recent molecular studies have identified EWSR1 or FUS gene rearrangements, particularly FUS: NFATC2 fusions, in a subset of SBCs, suggesting a neoplastic component in their pathogenesis [27, 28]. However, SBCs remain less cellular and less vascular than ABCs, and their histopathology is generally less complex [26].

While the precise cause is unknown, the dominant hypothesis attributes SBCs to a focal imbalance in skeletal development and intracystic fluid dynamics. It is hypothesized that an obstruction of intramedullary venous drainage leads to elevated intracystic pressure [29]. This increased pressure causes bone resorption by osteoclasts and stimulates the production of interstitial fluid, creating a self-perpetuating cycle of cyst expansion. Supporting this, analysis of SBC fluid has shown elevated levels of inflammatory mediators and enzymes, including prostaglandins and lysosomal enzymes, as well as increased concentrations of RANKL, which promotes osteoclast activity [30–32]. SBCs are clinically classified by their proximity to the physis, which predicts growth behavior and risk [33]. Active cysts lie within 1 cm of the growth plate, exhibit greater growth potential, and carry higher risks of pathological fracture and recurrence. As the bone elongates, the cyst migrates diaphyseally and becomes latent; these lesions have lower growth potential and are more likely to heal spontaneously, particularly as skeletal maturity nears [26]. Table 1 summarizes the key histopathological features of ABCs and SBCs.

Radiographic features

Plain radiography is the initial imaging modality of choice and often provides strong clues to the diagnosis. ABCs typically present as expansile, lytic lesions with well-defined margins, cortical thinning, and often a soap-bubble or multiloculated appearance due to internal septations [34]. They frequently cause bone expansion and may show periosteal reactions. Computed tomography (CT) can further delineate the extent of bone destruction, cortical breach, and internal septations; however, it is less sensitive than MRI in evaluating fluid characteristics and soft tissue involvement [35]. Magnetic resonance imaging (MRI) is particularly valuable for ABCs, as it often reveals multiple fluid-fluid levels within the cystic spaces, reflecting blood of different ages, and shows septal enhancement after contrast administration, features that are highly suggestive of ABC [36]. MRI also provides superior assessment of soft tissue extension and helps distinguish ABCs from malignant lesions [37, 38].

On the other hand, SBCs usually appear on plain radiographs as well-circumscribed, unilocular, radiolucent lesions with thin sclerotic margins, most commonly in the metaphysis of long bones [39]. They are often discovered incidentally or after a pathological fracture, which may result in a fallen fragment sign, a cortical bone fragment lying dependently within the cyst [40]. CT can help confirm the unilocular nature and assess cortical integrity. On MRI, SBCs typically show a single fluid-filled cavity [26].

Clinical course and natural history

ABCs typically manifest as swelling, pain, or pathological fracture, though some cases are discovered incidentally on imaging [18]. The clinical course of ABCs is unpredictable, with some lesions growing rapidly and causing substantial bone destruction, while others remain stable or regress spontaneously [18]. Recurrence after treatment is relatively common, especially in younger patients and when intralesional curettage alone is performed, with reported rates ranging from 10% to 30% depending on the location and completeness of excision [41]. Secondary ABCs, which arise in association with other bone lesions, tend to have a higher recurrence rate and may require more aggressive management [42].

SBCs may resolve spontaneously, particularly following a fracture; however, persistent or enlarging cysts can predispose to recurrent fractures or progressive bone deformity [4]. Compared with ABCs, recurrence after intervention is less frequent, and the overall prognosis is favorable, with most patients achieving complete recovery and subsequent bone remodeling over time [43].

Table 1 Histopathological and molecular comparison of ABCs and SBCs

Feature	Aneurysmal bone cyst (ABC)	Simple bone cyst (SBC)
Cystic spaces	Multiloculated, blood-filled, no true endothelial lining	Single cavity, fluid-filled, thin fibrous wall
Septa	Thick, fibrous, with giant cells, fibroblasts, woven bone	Thin, fibrous, sparse cellularity
Giant cells	Numerous osteoclast-type in septa	Scattered, rare
Bone formation	Reactive woven bone, osteoid, osteoblastic rimming	Occasional woven bone in wall
Vascularity	Highly vascular, blood-filled spaces	Poorly vascular, minimal hemorrhage
Molecular markers	USP6 gene rearrangement (primary ABC)	FUS/EWSR1-NFATC2 fusions in some cases

ABC, aneurysmal bone cyst; SBC, simple (unicameral) bone cyst.

Indications for interventions

Intervention for ABCs is indicated in cases of pain, rapid lesion expansion, risk of pathological fracture, and neurological symptoms [18]. Intralesional curettage, with or without adjuvant therapy, remains the standard approach; however, minimally invasive modalities such as percutaneous sclerotherapy, SAE, or systemic therapy are increasingly considered in surgically challenging locations [44]. Gross total resection is preferred when feasible, as incomplete excision is associated with higher recurrence rates [45].

In contrast, SBCs are often managed conservatively when asymptomatic, as many resolve spontaneously [4]. Surgical intervention is indicated in cases of persistent pain, risk of fracture, failure of healing after fracture, or when the lesion is large, expanding, or located in a weight-bearing bone where structural integrity is compromised [4]. Treatment options include intralesional curettage with or without bone grafting, elastic intramedullary nailing, and corticosteroid injection, with the choice determined by cyst size, anatomical location, and patient-specific factors [43].

Established and emerging treatments for ABCs and SBCs

The treatment of ABCs is guided by the lesion's size, location, and the patient's symptoms. The primary goals are to eradicate the lesion, prevent recurrence, restore bone integrity to prevent fracture, and preserve function [46]. The therapeutic landscape has evolved from exclusively open surgical approaches to include a variety of effective minimally invasive options.

Box 1. Treatment algorithm for ABCs.

Step 1. Confirm diagnosis and map anatomic risk

- Establish diagnosis and define lesion extent, proximity to the physis, and neurovascular/spinal risk using appropriate imaging and multidisciplinary review as indicated.
- Identify features that increase procedural morbidity (e.g., pelvis/spine, periarticular location, high vascularity) and prioritize limb/spine stability and function.

Step 2. Stratify by surgical accessibility and expected morbidity

- In low-morbidity, surgically accessible lesion (typical extremity locations), proceed with extended intralesional curettage (often with high-speed burr) and appropriate defect management.
- In anatomically complex/high-morbidity lesion (e.g., pelvis/spine/sacrum; high-risk periarticular sites), minimally invasive options or staged/adjunct approaches are preferred to reduce morbidity and improve safety.

Step 3. Choose primary local therapy (site- and morbidity-adapted)

- **Minimally invasive first-line (especially complex/high-risk sites):**
  - Sclerotherapy using an image-guided sclerosing agent, with expectation of possible repeat sessions and structured imaging response assessment.
  - Percutaneous steroid injection (± calcitonin) as an alternative minimally invasive strategy, particularly when open surgery is undesirable, mandate close surveillance for non-response.
- **Surgery-first (when feasible and function-preserving):**
  - Intralesional Curettage with adjuvants where local control is prioritized and anatomy permits; adjuvant selection should reflect site-specific risks and surgeon experience.

Step 4. Consider adjunctive SAE

- Use preoperative SAE to reduce bleeding risk and facilitate surgery in selected highly vascular or anatomically challenging lesions, and/or as part of a staged strategy where appropriate.

Step 5. Escalation and salvage strategy

- Incomplete response or recurrence after minimally invasive therapy:
  Repeat minimally invasive treatment when appropriate or escalate to extended curettage ± adjuvants based on anatomy and functional considerations.
- Refractory disease, selected expendable bones, or scenarios where maximal local control outweighs morbidity:
  Consider en bloc resection with reconstruction as needed; reserve for carefully selected cases given technical complexity and potential functional compromise.

Step 6. Follow-up and decision triggers

- Implement standardized follow-up imaging and clinical assessment.
- Define early triggers for escalation (persistent symptoms, radiographic non-response/progression, recurrent disease), particularly in pelvis/spine and periarticular sites.

Table 2 Clinical outcomes of intralesional curettage and en bloc resection for ABCs

Study ID	Study design	Sample size	Included patients	Treatment used	Efficacy	Recurrence	Side effects
Intralesional curettage with or without graft or adjuvant therapy							
Deventer et al., (2021) [48]	Single-center retrospective	74	Primary ABC	Intralesional curettage ($n=34$)	N/A	47.1%	Wound healing disorder (5.9%)
Grahneis et al., (2019) [55]	Single-center retrospective	65	Primary ABC	Curettage alone ($n=33$)	64% healed	27%	N/A
				Curettage with phenol ($n=21$)	76% healed	9.5%	
Keçeci et al., (2014) [63]	Single-center retrospective	76	Primary ABC	Curettage alone ($n=14$)	N/A	7.1%	N/A
				Curettage with high-speed burr ($n=19$)	N/A	15.8%	
				Curettage + high-speed burr + alcohol/phenol ($n=43$)	N/A	14%	
Myilvahanan et al., (2023) [64]	Case report	2	ABC of Talus	Extended curettage + Graft	100% healed	0%	N/A
Crowe et al., (2015) [65]	Case-series	11	Hand/Wrist/ Forearm ABC	Curettage + Burring ± adjuvant ($n=10$)	90% healed	10%	Revision needed in one patient
Yenigül et al., (2022) [50]	Single-center retrospective	172 (88 ABC)	Locally aggressive benign tumors	Extended curettage (no adjuvant)	88.63% healed	11.36%	Infection ($n=4$), fracture ($n=1$), and Osteoarthritis ($n=1$)
Jamshidi et al., (2019) [60]	Single-center retrospective	17	Scapular ABC	Extended curettage ($n=12$)	91.67% healed	8.33%	Functional impairment
Wang et al., (2014) [66]	Single-center retrospective	31	Primary ABC	Curettage + Burring + Graft	96.8% healed	3.2%	Growth deformity, collapse
Cummings et al., (2010) [68]	Single-center retrospective	29	ABC	Curettage + Argon beam ($n=17$)	100% healed	0%	N/A
				Curettage with or without phenol ($n=12$)	66.67%	33.33%	
En bloc resection							
Sami et al., (2024) [59]	Single-center retrospective	19	Metatarsal ABC	En bloc resection + Allograft	100% healed	0%	Infection, non-union, fracture
Deventer et al., (2021) [48]	Single-center retrospective	74 (En Bloc Resection = 8)	Primary ABC with a large soft tissue component in the pelvis, femur, and a massive manifestation in the spine	En bloc resection	75% healed	25%	Fracture ($n=2$), healing disorder ($n=1$) and spondylolisthesis ($n=1$)
Grahneis et al., (2019) [55]	Single-center retrospective	65	ABC of Soft tissues	En bloc resection ($n=4$)	100% healed	0%	N/A
Jamshidi et al., (2019) [60]	Single-center retrospective	17	Scapular ABC	En bloc resection ($n=5$)	100% healed	0%	Functional impairment
Keçeci et al., (2014) [63]	Single-center retrospective	76	Primary ABC	Resection ($n=9$)	N/A	0%	N/A
Sclerotherapy							
Rahul et al., (2018)[69]	Retrospective review	15	ABC diagnosed by radiology with biopsy confirmation	Percutaneous sclerotherapy with 3% polidocanol (30 mg/mL) under fluoroscopy/CT + GA; 3 injections at 6-week intervals	Pain resolved (VAS = 0) in 14/15 (93.33%)	0%	Minor local inflammatory reactions in 2 patients

ABC, aneurysmal bone cyst; N/A, not available

Established treatments of ABCs

Surgical interventions of ABCs

Intralesional curettage: Intralesional curettage has historically been the standard of care for ABCs [47]. The procedure involves creating a cortical window, removing the cystic contents and membranous lining, and often abrading the cavity with a high-speed burr to eliminate microscopic disease [5]. However, curettage alone is associated with substantial and heterogeneous recurrence across reports [7], with individual series describing recurrence rates in the 30–40% range [47–49]. Furthermore, a systematic review by Bavan et al., analyzed 17 studies involving 594 patients, reported an aggregate failure or recurrence rate of 14.4% for curettage (with or without adjuvants), with rates varying from 0% to 40% across different studies [6]. Nevertheless, functional outcomes are usually favorable, with most patients returning to normal

Table 3 Comparative effectiveness and recurrence across treatment modalities for ABCs

Study ID	Study design	Sample size	Included patients	Treatment used	Efficacy	Recurrence	Side effects
Strohm et al., (2023) [51]	Systematic review & Meta-analysis	163 studies	ABCs	Curettage alone, Curettage + Graft, En bloc, Adjuvants	Curettage alone: 91% healing, Curettage + Graft: 96%, En Bloc: 98%	Curettage alone: 22%, Curettage + Graft: 15%, En Bloc: 6%	N/A
Bavan et al., (2021) [6]	Systematic review	28 studies ($n=594$ curettage, $n=387$ injection)	Primary ABC (long bones)	Curettage (± adjuvant), En bloc, Sclerotherapy	Curettage: 85.6% success	Curettage: 14.4%	N/A
Cruz et al., (2021) [93]	Meta-analysis	13 studies ($n=416$)	Primary ABC	Percutaneous (Sclerotherapy, Embolization)	Sclerotherapy: 94% success	Sclerotherapy: 6%, Embolization: 19%	N/A
Di Costa et al., (2024) [62]	Systematic review	37 studies ($n=140$)	ABC with fracture	Curettage, En bloc, Adjuvants	Curettage: 88.6% healing	N/A	N/A
Parker et al., (2019) [94]	Systematic review	21 studies ($n=272$)	Spinal ABC	Curettage, En bloc, Adjuvants	En Bloc: high cure, Curettage: moderate	Curettage: 25%, En Bloc: lower	Neurologic deficit, deformity
Döring et al., (2021) [47]	Large retrospective	90	Primary ABC	Curettage (± burring, phenol), En bloc	N/A	Curettage: 33%, En Bloc: lower	N/A
Gettleman et al., (2024) [95]	Large retrospective	129	Pediatric ABC	Curettage (± adjuvant)	N/A	Curettage + Adjuvant: 25.3%, Curettage without adjuvant: 23.7%	N/A
Levanon et al., (2023) [96]	Large retrospective	130	ABC	Curettage ± Cryoablation	N/A	Curettage + Cryo: 7.1%, Curettage without Chryo: 9.1%	N/A
Erol et al., (2015) [97]	Retrospective	64	Pediatric ABC	Extended curettage (burr, cautery, graft)	95% MSTS score	7%	N/A
Puthoor et al., (2021) [98]	Prospective comparative	48	ABC	Sclerotherapy vs. Curettage + Graft	Sclerotherapy: 100%, Curettage: 82%	Curettage: 18%	Sclerotherapy: mild, Curettage: more
Dormans et al., (2004) [99]	Retrospective	45	Pediatric ABC	Curettage (4-step), En bloc	82% success	N/A	N/A
Palmisciano et al., (2022) [42]	Systematic review	80 studies ($n=220$)	Adult Spinal ABC	Curettage, En bloc, Adjuvants	91.8% improvement	Secondary ABC: 19.2%	N/A

Table 3 (continued)

Study ID	Study design	Sample size	Included patients	Treatment used	Efficacy	Recurrence	Side effects
Yang et al., (2013) [100]	Retrospective cohort	48	Children/adolescents with ABC	Steroid injection (methylprednisolone)	68% cure	N/A	Minimal, minor complications
Samargandi et al., (2023) [10]	Systematic review/meta-analysis	729	Primary ABCs	Sclerotherapy (various agents), calcitonin+steroid	74.3% complete healing	3.5%	Doxycycline: minimal, Ethibloc/embolization: higher
Chang et al., (2017) [71]	Retrospective case series	9	ABCs (various sites)	Calcitonin + steroid injection	75% complete relief	25% (4/8)	N/A
Masthoff et al., (2021) [101]	Retrospective cohort	16	Refractory/nonresectable ABCs	Embolization + sclerotherapy	100% partial/complete response	0%	N/A
Aiba et al., (2018) [91]	Retrospective	30	ABC	Endoscopic curettage	90% healed	10%	Nerve palsy (transient)
Varshney et al., (2010) [11]	Prospective comparative	45	Previously untreated ABCs	Polidocanol sclerotherapy	93.3% healed	4.44%	Local induration 82.2% and hypopigmentation 24.4%
		46		Extended curettage/intralesional excision + bone grafting	84.8% healed	15.22%	Deep infections (3 patients) and superficial infections (5 patients)

activities. Complications, though uncommon, can include growth plate injury and deformity [50].

Intralesional curettage with adjuvant therapy: The addition of adjuvant therapies to curettage aims to improve local control and reduce recurrence. The rationale is to destroy any residual neoplastic cells left behind on the cavity walls, effectively sterilizing the surgical bed [5]. Meta-analyses highlight that curettage combined with autologous cancellous bone grafting yields better healing and lower recurrence than curettage alone [51]. Other adjuvants, including high-speed burring, penalization, cryotherapy, argon beam coagulation, and bone cement, have been associated with improved local control in selected series, although reported effects vary across studies [52–54]. Preoperative SAE has also been shown to further decrease recurrence risk [55]. Overall complication rates are generally low, but modality-specific risks should be considered during patient selection and surgical planning [56].

En bloc resection: En bloc resection, which entails complete removal of the cyst with a margin of healthy bone, provides the highest cure rates for ABCs, with recurrence rates approaching zero in many series [57]. This radical approach is most suitable for lesions in surgically accessible and expendable bones such as the fibula, clavicle, or ribs, where resection causes minimal functional loss [6, 57]. However, its application in weight-bearing bones is limited due to the high surgical morbidity, need for complex reconstructions, and risk of functional compromise [58]. Across published series and systematic reviews, resection consistently demonstrates superior local control versus extended curettage, albeit with the trade-off of greater technical complexity and possible functional impact in selected locations [59–62] (Table 2).

Minimally invasive and percutaneous interventions for ABCs

Percutaneous steroid injection: Percutaneous steroid injection, often combined with calcitonin, has emerged as a minimally invasive and effective treatment for ABCs. Several retrospective and prospective series showed that patients achieve meaningful clinical improvement and radiographic healing, although non-responders appear at higher risk of subsequent recurrence and therefore warrant close monitoring [70–72]. A meta-analysis of 25 studies involving 729 patients showed that calcitonin plus methylprednisolone had comparable efficacy and safety to other sclerotherapy agents, with a low recurrence rate of 3.5% [10]. Collectively, these findings support steroid-based injections as a strong option for ABCs.

Percutaneous bone marrow injection: Percutaneous bone marrow or allogenic graft injection is a minimally invasive technique designed to stimulate healing in ABCs by introducing osteogenic cells and scaffolding materials directly into the cystic cavity [73]. This approach leverages the regenerative capacity of autologous bone marrow, which is rich in mesenchymal stem cells, and combines it with adjuncts such as demineralized bone matrix (DBM), platelet-rich fibrin (PRF), or allogenic bone particles to promote ossification and cyst resolution [74]. Clinical evidence supports its utility, with comparative studies showing up to 80% healing rates after a single injection of bone marrow concentrate combined with DBM and PRF, outcomes comparable to curettage with bone grafting [75]. Case series have reported similar results, with healing rates around 79% for percutaneous DBM and bone marrow injections [76]. Collectively, these findings indicate that percutaneous bone marrow or allogenic graft injection is a safe, effective, and repeatable treatment option that provides a valuable alternative when open surgery is undesirable or in high-risk settings.

Injectable bone substitutes: Injection of bone substitutes such as calcium sulfate, calcium phosphate, or biphasic ceramic materials has emerged as a minimally invasive option for treating ABCs [77]. The rationale is to provide immediate mechanical stability, fill the cystic cavity with an osteoconductive scaffold, and promote gradual replacement by native bone while minimizing long-term complications. Clinical evidence supports this approach, with case reports and series showing encouraging results. A reported case of a C2 ABC treated with a single biphasic ceramic injection achieved complete ossification, cortical remodeling, and restoration of spinal canal anatomy within six months, without complications [78]. These findings suggest that bone substitute injection is a safe, effective, and repeatable strategy that can achieve durable cyst healing and functional recovery.

Sclerotherapy: Sclerotherapy is a minimally invasive technique for treating ABCs that involves percutaneous injection of sclerosing agents directly into the cyst cavity under imaging guidance [79, 80]. The mechanism is based on inducing endothelial damage, thrombosis, and fibrosis within the lesion, which ultimately promotes ossification and healing [79]. This approach is particularly valuable for anatomically complex or surgically high-risk sites, including the pelvis and spine [10]. Clinical evidence consistently demonstrates its efficacy, with prospective and retrospective studies showing high healing rates, rapid pain relief, and functional recovery. For example, a polidocanol-based study in 26 patients achieved 92.3% ossification and a substantial pain reduction, with only two recurrences successfully managed by a repeat injection [81]. Larger series have reported disease control rates of 96–100% after two to four injections, with minimal complications such as transient nausea, ecchymosis, or local discomfort [82, 83].

Selective arterial embolization (SAE): SAE is a minimally invasive technique increasingly used in the management of ABCs, especially in highly vascular sites such as the spine, pelvis, and sacrum [35]. The procedure involves angiographic identification of feeding arteries followed by targeted occlusion with embolic agents. By devascularizing the lesion, SAE induces thrombosis and promotes cyst regression and ossification, reducing pain, controlling growth, and lowering the risk of intraoperative bleeding if later surgery is required [84, 85]. In a cohort of 36 patients, effectiveness reached 94% with only minor complications such as transient paresis or limited skin necrosis [86]. Another series of 18 patients reported an 88.9% response rate, no major complications, and only two recurrences managed surgically [85] (Table 3).

Radiofrequency ablation (RFA) and cryoablation: RFA and cryoablation are minimally invasive, image-guided thermal ablation techniques that have gained increasing use in the management of ABCs [87]. RFA uses high-frequency alternating current to generate heat and induce coagulative necrosis within the lesion, while cryoablation uses rapid freezing, typically with argon gas, to form an ice ball that destroys tumor cells [88]. Both procedures are performed percutaneously under CT or MRI guidance, ensuring precise targeting and sparing adjacent tissues. Their primary aims are local tumor control, pain reduction, and functional recovery, achieved with lower morbidity than open surgery. Clinical studies have shown encouraging results. In a cohort of 20 patients managed with RFA (with or without cementation), pain scores dropped dramatically from a mean VAS of 8.4 to 0.2, with no recurrences or complications after one year of follow-up [87]. Cryoablation, used either alone or with cementoplasty, has similarly demonstrated high rates of radiological healing and symptom resolution in both pediatric and adult populations, including spinal and recurrent cases [89, 90].

Endoscopic curettage: Minimally invasive curettage via endoscopy is an emerging approach for managing ABCs, designed to lower surgical morbidity while ensuring effective local control. Using arthroscopic or endoscopic guidance, surgeons access the cyst through small incisions, enabling direct visualization and precise curettage until healthy bone is exposed [91]. This method reduces blind spots, prevents unnecessary bone loss, and helps preserve vital structures such as the physis in pediatric patients. Compared with open surgery, the technique causes less soft tissue disruption, carries a lower risk of infection, and yields improved cosmetic outcomes. The defect is typically filled with autograft, allograft, or synthetic bone substitutes to promote consolidation [92]. A series of 30 patients demonstrated a 10% recurrence rate, solid union at an average of 3.2 months, with excellent functional recovery [91].

Established treatments of SBC

A pragmatic, fracture-risk–centered treatment pathway for simple bone cysts is summarized in Box 2.

Box 2. Treatment algorithm for simple bone cysts.

Step 1. Confirm diagnosis and exclude mimics

- Typical radiographic appearance in a skeletally immature patient is often sufficient; obtain MRI/CT or specialist review when features are atypical (aggressive periosteal reaction, soft-tissue mass, unusual location) or when diagnosis is uncertain.

Step 2. Stratify by clinical scenario and fracture risk
Assess:

- **Symptoms** (pain, functional limitation)
- **Location** (proximal humerus vs proximal femur or other weight-bearing bones)
- **Activity** (lesion close to physis "active" vs migrating away "latent")
- **Structural risk** (large cyst volume, cortical thinning, high-risk geometry)

Step 3. Low-risk pathway

- Proceed with observation and periodic radiographs in asymptomatic lesions with low fracture risk (often latent), particularly as many stabilize or resolve with skeletal maturity.

Step 4. Intervention pathway (symptomatic and/or higher fracture risk)

- **Minimally invasive, bone-preserving options (typical first-line):**
  - Percutaneous decompression (e.g., cannulated screw or similar technique to reduce intracystic pressure) with or without graft/substitute.
  - Percutaneous injection-based therapy (commonly corticosteroid injection, bone marrow aspirate and/or bone substitutes), recognising that repeat sessions may be required.
- **High-risk location or weight-bearing bone (especially proximal femur):**
  - Consider prophylactic internal fixation (often combined with decompression and/or graft/substitute) when mechanical risk is substantial, to reduce the consequence of fracture and permit safer healing.

Step 5. Pathological fracture pathway

- Treat the fracture per standard orthopaedic principles; fracture healing can contribute to cyst resolution in some cases.
- After fracture union (or during fixation when indicated), consider adjuvant cyst-directed treatment (decompression/injection/grafting) if the cavity persists or remains high risk.

Step 6. Escalation and salvage

- Escalate to surgical curettage and grafting/substitute (typically with internal fixation when stability is a concern, particularly in weight-bearing sites) in cases of persistent cyst, repeated recurrence/non-healing after percutaneous therapy, or progressive structural compromise.

Step 7. Follow-up and triggers for escalation

- Standardize follow-up imaging and define triggers (worsening pain, enlargement, persistent cortical thinning, new fracture, or failure to show interval healing) that prompt repeat percutaneous therapy or surgical escalation.

Table 4 Comparative outcomes of treatments for SBCs

Study ID	Study design	Sample size	Included patients	Treatment used	Efficacy	Recurrence	Side effects
Zhang et al., (2019) [116]	Retrospective cohort	18	Children with humeral SBCs	ESIN + steroid injection	100% full/partial healing	0%	Nail irritation ($n=1$), valgus deformity ($n=1$)
Maximen et al., (2024) [123]	Multicenter retrospective	201	Children with proximal femoral SBCs	ESIN ± injection/grafting	67–79% healing (first op)	N/A	Limb length discrepancy
Masry et al., (2021) [124]	Retrospective case series	37	Children with SBCs	ESIN decompression	78% Capanna grade 1 healing	2 recurrences	N/A
		34		Curettage + bone graft	85% Capanna grade 1 healing	1 recurrence	N/A
Li et al., (2016) [125]	Retrospective comparative	46	Children with SBCs	TEN (intramedullary nailing)	69.6% complete healing	13%	Skin irritation ($n=3$), fractures ($n=2$)
		23		BMAC injection	60.9% complete healing	13%	4 fractures
Mylle et al., (1992) [104]	Retrospective	21	SBCs (various sites)	Curettage ± bone graft	43% favorable	29%	N/A
		29		Intralesional steroid injection	90% favorable	5%	N/A
Bishay, (2015) [126]	Case series	12	Capitate SBCs	Curettage (no graft)	100% union	0%	2 pain after activity
Canavese et al., (2011) [127]	Retrospective	46	Children/adolescents with SBCs	Percutaneous curettage	70% satisfactory healing	N/A	N/A
		46		BMAC injection	21% satisfactory healing	N/A	N/A
Hashemi-Nejad & Cole, (1997) [128]	Retrospective	32	Children with SBCs	Intralesional steroid injection	41% satisfactory healing (1st injection)	High in weight-bearing bones	N/A
Da Costa Mendes et al., (2020) [103]	Case report	1	Mandibular SBC	Intralesional steroid injection	Bone formation at 6 mo	0%	None
Wright et al., (2008) [129]	RCT	90	Children with SBCs	Steroid vs. BMAC injection	Steroid: 42% healing, BMAC: 23%	Not specified	No major
D'amato et al., (2020) [130]	Retrospective	24	SBCs	BMAC + EDDBM injection	83.3% success	N/A	N/A

Conservative management

Conservative therapy for SBCs typically involves observation or minimally invasive interventions, especially in cases with non-displaced pathological fractures or asymptomatic lesions. Studies showed that most SBCs, particularly those in the upper extremity, can be managed conservatively, with spontaneous healing observed in many cases [58]. Conservative approaches may include immobilization and close radiographic follow-up, as the majority of pathological fractures treated by this approach heal within six weeks, and secondary deformities are rare [102]. Intralesional steroid injections have also been used as a conservative, low-cost technique, promoting bone formation and preserving vital structures, with documented efficacy in long bone SBCs [103, 104].

Surgical management of SBCs

Intralesional curettage: Intralesional curettage is the standard surgical treatment for SBCs, aiming to remove the cyst lining and contents to stimulate bone healing. Clinical outcomes of intralesional curettage for SBCs and other benign osseous lesions demonstrate relatively high recurrence rates, with studies reporting 27–51% recurrence and some series noting a median of 47% [105, 106], largely influenced by the adequacy of cyst wall removal [107]. Greater benefits have been observed with the use of adjuvant therapies such as high-speed burring, cryotherapy, polymethyl methacrylate (PMMA) cement, or chemical agents. Meta-analyses report that recurrence rates decrease to approximately 11.9% with high-speed burring and around 20% with PMMA, although the efficacy of chemical agents such as phenol or hydrogen peroxide remains inconsistent [108, 109]. Overall, meticulous curettage remains the critical determinant of success, with grafting providing structural support and adjuvants offering additional protection against recurrence, particularly in aggressive lesions.

Cyst decompression and elastic stable intramedullary nailing (ESIN): Cyst decompression techniques for SBCs

Table 5 Summary of clinical experience of denosumab for ABCs

Study ID	Study design	Sample size	Included patients	Treatment used	Efficacy	Recurrence	Side effects
Evangelisti et al., (2024) [134]	Retrospective cohort	6	Relapsed, symptomatic spinal ABCs	Denosumab (120 mg, standard GCTB protocol)	All improved clinically/radiologically; long-term disease control	3/6 relapsed (responded to re-challenge or MSCs)	Non-significant
Mohamed et al., (2023) [131]	Systematic review	13 studies, 40 patients	ABCs (various sites)	Denosumab	94.7% clinical response	Relapse possible after cessation	Minimal complications, no deaths
Dürr et al., (2019) [142]	Retrospective case series	6	ABCs (sacrum, pelvis, extremities)	Denosumab (120 mg, GCTB protocol)	4/6 stable long-term; 2 recurred (1 re-treated)	2/6 recurred after cessation	Severe hypercalcemia in 1 child
Masry et al., (2024) [136]	Retrospective cohort	20	Pelvic ABCs	Denosumab (solo, 6–9 doses)	80% disease control, MSTS 92%	0% at mean 38.5 mo follow-up	15% mild (GI upset, bone pain)
Palmerini et al., (2018) [14]	Case series	9	ABCs (spine, pelvis, long bones)	Denosumab (median 8 doses)	All had pain relief, bone formation	0% at median 23 mo	Negligible
Skubitz et al., (2015) [143]	Case report	1	Large sacral ABC	Denosumab (GCTB protocol)	Pain resolved, bone formation, no recurrence at 12 mo	0% at 12 mo	Non-significant

ABC, aneurysmal bone cyst; GCTB, giant cell tumor of bone; MSTS, Musculoskeletal Tumor Society score; MSCs, mesenchymal stromal cells; GI, gastrointestinal.

primarily involve creating a channel between the cyst cavity and the medullary canal to relieve intracystic pressure and promote healing [110]. Methods such as percutaneous drilling or decompression with cannulated pins have demonstrated high healing rates, with some studies reporting successful outcomes in up to 89.5% of cases with low recurrence rates [111]. These minimally invasive approaches allow for early mobilization, reduced surgical trauma, and preservation of bone integrity, making them suitable for pediatric patients. However, although cyst decompression alone can be effective, some patients may experience partial healing or residual cysts, and recurrence risk tends to be higher in younger children or in lesions located near the growth plate [112].

Elastic stable intramedullary nailing (ESIN) is a widely used decompression technique that provides both continuous drainage of cystic fluid and mechanical stability [113]. Studies consistently show that ESIN leads to high rates of complete or partial radiographic healing, ranging from 66% to 100%, with very low recurrence and complication rates. ESIN is particularly advantageous in cases with pathological fractures, as it stabilizes the bone and allows for early return to activity [114–116]. Functional outcomes are excellent, with most patients regaining full range of motion and experiencing minimal pain postoperatively. Complications are rare and generally mild, such as nail irritation or minor deformity, and the need for reoperation is low [117, 118]. Other adjunctive techniques, such as combining ESIN with steroid injections, bone grafting, or demineralized bone matrix, have also been explored. ESIN combined with intralesional steroid injection has shown 100% complete or partial healing with no recurrences in some series, while ESIN with bone grafting or bone marrow injection also yields excellent radiological and functional outcomes [119, 120]. Overall, the addition of adjuvant therapies to decompression techniques can optimize outcomes, particularly in complex or recurrent SBCs (Table 4).

Intralesional injections

Intralesional corticosteroid injection for SBCs typically results in rapid pain relief and functional improvement, with significant reductions in pain scores observed as early as 6 weeks post-injection. In musculoskeletal applications, corticosteroid injections have been shown to reduce pain scores by up to 50% at 6 weeks and 3 months, with the effect often diminishing by 6 months [121, 122]. However, recurrence rates for SBCs treated with corticosteroids alone are reported to range from 30% to 50% at 1–2 years, indicating that while initial symptomatic control is excellent, long-term cyst resolution is less reliable. Complications are rare, with most studies reporting no serious adverse events, but minor risks include local tissue atrophy or infection [121, 122].

Emerging therapies for ABCs and SBCs

Denosumab (RANKL inhibition)

Denosumab is a recent pharmacologic advance for ABCs and may be considered in cases where conventional surgical or interventional options are inadequate [131]. It is a fully human monoclonal antibody that binds RANKL with high affinity and specificity, thereby inhibiting a key pathway that drives osteoclast differentiation, activation, and survival [132, 133]. By blocking RANKL–RANK signaling, denosumab suppresses osteoclast-mediated bone resorption, limits progressive bone destruction, and can

promote structural stabilization of the lesion [132, 133]. This mechanism is biologically relevant in ABCs, which are enriched in osteoclast-like giant cells and exhibit heightened RANK/RANKL pathway activity contributing to osteolysis and aggressive behavior [3].

Denosumab is primarily used as a second-line or adjunctive option for surgically challenging ABCs, including unresectable lesions, recurrent disease refractory to conventional interventions, and tumors in anatomically complex/high-risk locations where surgery would carry substantial morbidity [134]. It is particularly reported in spine and pelvis lesions, where it may relieve pain, improve neurological deficits, and stabilize the lesion while reducing the need for high-risk surgery and associated complications such as hemorrhage [135, 136].

Clinical outcomes and challenges: The efficacy of denosumab is often rapid and profound. Studies consistently report high rates of positive clinical response, including significant pain relief and functional improvement within weeks of treatment initiation [15, 137, 138]. Radiologically, this corresponds to progressive ossification, formation of a sclerotic rim, and a reduction in the cystic components of the lesion. Beyond its role as primary therapy, denosumab is also valuable as a neoadjuvant agent. Preoperative administration can downstage large, aggressive tumors, reduce their vascularity, and induce a sclerotic shell, thereby facilitating a safer, less extensive, and more effective surgical resection.

Despite demonstrated effectiveness, denosumab use in ABCs, particularly in pediatric patients, raises important safety and durability concerns, including recurrence after treatment discontinuation and the need for re-treatment in some cases [139]. A key risk in skeletally immature patients is severe rebound hypercalcemia after cessation, likely reflecting a surge in osteoclast activity and higher baseline bone turnover, necessitating prolonged post-treatment calcium monitoring [67, 140]. Additional uncertainties relate to potential long-term effects on the growing skeleton, including reports of sclerotic metaphyseal bands and possible impacts on bone modelling and linear growth, although long-term consequences remain incompletely defined [141].

Role of denosumab in SBCs: In contrast to its role in ABCs, there is currently no substantive evidence or established biological rationale to support the use of denosumab for the treatment of SBCs. The pathophysiology of SBCs is believed to be related to a localized disturbance in venous drainage, leading to fluid accumulation and pressure-induced bone resorption, rather than a neoplastic process driven by osteoclast-like giant cells. The cellular and molecular environment of an SBC does not involve the RANKL-mediated osteoclastogenesis that denosumab targets. Therefore, its use is not supported by existing literature and is not a recommended treatment modality for this condition (Table 5).

Special considerations and management challenges

Although a wide range of treatment modalities exists for SBCs and ABCs, their application is not uniform. Management decisions must be individualized, taking into account patient-specific factors and the biological and anatomical characteristics of each lesion. Pediatric patients, lesions situated in complex anatomical regions, and cases characterized by recurrence each present distinct clinical challenges. These scenarios require carefully tailored therapeutic strategies.

Pediatric patients

The vast majority of ABCs and SBCs are diagnosed in children and adolescents, a demographic that introduces several layers of complexity to treatment planning.

Higher recurrence rates

Younger age is one of the most consistently reported independent risk factors for treatment failure and lesion recurrence for both cyst types. For SBCs, patients younger than 10 years have a significantly higher risk of recurrence following treatment [144]. Similarly, for ABCs, age below 10 has been associated with lower recurrence-free survival after surgical treatment [47].

Growth plate (Physeal) proximity

Many of these lesions arise in the metaphysis, immediately adjacent to the physis (growth plate) [19]. This proximity presents a dual risk. First, an aggressive, expansile lesion like an ABC can directly damage or disrupt the growth plate, leading to premature closure, angular deformities, or limb length discrepancy [145]. Second, the treatment itself can cause iatrogenic physeal injury. Aggressive intralesional curettage, or the use of chemical or thermal adjuvants, can damage the delicate chondrocytes of the growth plate if not applied with extreme caution [107, 146].

Anatomically challenging locations

Anatomical site largely dictates the risk and feasibility of managing bone cysts, with the spine, pelvis, and skull base being particularly challenging. Spinal ABCs, which account for approximately 15% of cases [147], threaten neural structures and stability, making surgery high-risk, with potential for dural tears, neurological injury, and spinal deformity [148]. Management often combines SAE, denosumab, and, when unavoidable, complex instrumented fusion [149]. Pelvic ABCs are difficult to access and carry major hemorrhagic and neurovascular risks [145], so minimally invasive approaches, SAE,

percutaneous sclerotherapy, and denosumab are commonly favored to avoid open surgery [136].

Management of recurrent lesions

Recurrence represents the most common form of treatment failure for both ABC and SBC [18]. The management of a recurrent lesion often requires a step-up in therapeutic aggressiveness [63]. The initial choice of treatment, patient age, and lesion location all influence the approach to recurrence [150]. Options include repeating the initial therapy, escalating to a more invasive procedure, or switching to an entirely different modality [6]. For SBC that recurs after steroid injection, a surgeon might proceed to curettage and bone grafting [151]. For an ABC that recurs after simple curettage, a repeat procedure with the addition of a potent adjuvant may be indicated to achieve better local control [55]. In cases where recurrence occurs in a location where further surgery is unfeasible or would be too morbid, alternative modalities become critical [14, 152].

Unmet needs and critical gaps in the management of ABCs and SBCs

From our perspective, management of ABCs and SBCs is limited chiefly by two gaps: diagnostic certainty and high-quality comparative evidence. In terms of diagnosis, distinguishing primary ABC from secondary ABC remains imperfect in routine practice; while USP6 rearrangements strongly support primary ABC, molecular testing is not universally available, and fusion partners are expanding, complicating test selection and interpretation [18, 153]. Imaging-based criteria for activity/healing also vary, which undermines trial comparability and day-to-day decisions such as when to stop sclerotherapy or resume sport. For SBCs, even basic questions (natural history near the physis, true fracture risk, and the definition of healed) lack consensus; legacy radiographic scores and heterogeneous endpoints limit meta-analyses and practical guidance [26]. Evidence quality is another core unmet need. For ABCs, most data on SAE, sclerotherapy agents, and denosumab come from small series, single-center cohorts, and systematic reviews that highlight promise but also underscore ongoing uncertainty regarding long-term oncologic durability and the risk of relapse following treatment discontinuation [154]. In pediatric patients, the role of denosumab remains particularly uncertain. Reports describe favorable responses in inoperable spinal and pelvic lesions but also document episodes of rebound hypercalcemia. These findings underscore the absence of standardized dosing or tapering protocols and the lack of long-term safety data [140, 155]. For SBCs, decades of clinical practice continue to rely on small or methodologically limited studies comparing treatments such as steroid injection, decompression, curettage with bone grafting, intramedullary nailing, and sclerotherapy. Even the sole Level I trial was modest in size, and durable superiority of any single modality, or treatment sequence, remains unproven [156].

Management and follow-up pathways remain highly fragmented. In anatomically complex ABCs involving the spine, pelvis, or skull base, a careful balance must be maintained between the risks of bleeding, neurological compromise, and recurrence against procedure-related morbidity. SAE and other percutaneous techniques are being adopted with increasing frequency; however, prospective treatment algorithms and head-to-head comparisons with surgical approaches remain scarce, particularly for pelvic lesions, where complication and reoperation rates continue to be substantially high [152]. For SBCs, unanswered questions include when to intervene versus observe; how to individualize care by age, cyst activity, and proximity to the physis; as well as whether to prioritize single-stage definitive surgery or staged minimally invasive care [26]. Across both entities, there is a need for: core outcome sets that integrate pain, function, time-to-healing, growth-plate complications, and return-to-activity; standardized imaging follow-up schedules and healing criteria; comparative-effectiveness trials; and validated risk tools using age, location, genetic markers, and cyst biomechanics to predict fracture and recurrence and to guide sequencing of treatments.

Future directions

Future studies should move beyond isolated case series toward prospectively designed programs that integrate biology, biomechanics, and precision intervention. Molecular stratification, quantitative imaging phenotypes, and biomechanical risk scoring could be combined to personalize therapy sequencing and to proactively prevent deformity in juxta-physeal lesions.

Technique innovation should prioritize growth-plate–sparing approaches, and smarter percutaneous regimens. In parallel, pragmatic, multi-center platform trials can test modular care bundles (e.g., SAE followed by sclerotherapy vs. sclerotherapy-alone vs. denosumab-bridging) with shared core outcomes and time-to-function endpoints, while embedded health-economics analyses clarify value in real systems.

Conclusions

Management of ABCs and SBCs has evolved toward risk-adapted strategies that balance local control against morbidity, growth-plate preservation, and functional outcomes, although most comparative evidence remains low-level and heterogeneous. For ABCs, an anatomy- and morbidity-stratified pathway is most actionable: surgically accessible lesions can be managed with extended intralesional curettage and appropriate reconstruction,

whereas anatomically complex or high-morbidity sites are well suited to minimally invasive first-line options such as image-guided sclerotherapy, with SAE used as definitive or adjunct therapy to reduce vascularity and facilitate safer intervention; escalation to en bloc resection is reserved for carefully selected refractory cases. Denosumab should remain a reserved option for aggressive, recurrent, or inoperable ABCs, with structured safety monitoring.

For SBCs, a fracture-risk–centered approach is recommended: asymptomatic, low-risk lesions may be observed, whereas symptomatic or higher-risk cysts warrant bone-preserving percutaneous strategies. In mechanically vulnerable locations, particularly the proximal femur, prophylactic stabilization should be considered to mitigate fracture consequence and support healing, with curettage and grafting/substitute (with fixation when needed) reserved for persistent lesions, repeated failure of percutaneous management, or progressive structural compromise.

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References

1. Mascard E, Gomez-Brouchet A, Lambot K. Bone cysts: Unicameral and aneurysmal bone cyst. Orthop Traumatol Surg Res. 2015;101:S119–27. https://doi.org/10.1016/j.otsr.2014.06.031.
2. Rosenblatt J, Koder A. Understanding unicameral and aneurysmal bone cysts. Pediatr Rev. 2019;40:51–9. https://doi.org/10.1542/pir.2015-0128.
3. Restrepo R, Zahrah D, Pelaez L, Temple HT, Murakami JW. Update on aneurysmal bone cyst: pathophysiology, histology, imaging and treatment. Pediatr Radiol. 2022;52:1601–14. https://doi.org/10.1007/s00247-022-05396-6.
4. Deventer N, Deventer N, Gosheger G, de Vaal M, Vogt B, Budny T. Current strategies for the treatment of solitary and aneurysmal bone cysts: A review of the literature. J Bone Oncol. 2021;30:100384. https://doi.org/10.1016/j.jbo.2021.100384.
5. van Geloven TPG, van de Sande MAJ, van der Heijden L. The treatment of aneurysmal bone cysts. Curr Opin Pediatr. 2023;35:131–7. https://doi.org/10.1097/MOP.0000000000001205.
6. Bavan L, Wijendra A, Kothari A. Efficacy of treatment interventions for primary aneurysmal bone cysts: a systematic review. Bone Jt open. 2021;2:125–33. https://doi.org/10.1302/2633-1462.22.BJO-2020-0168.
7. Sakai K, Minabe M, Hata K, Kamemoto K, Masuda K, Hashimoto K, et al. Successful treatment of a rapidly enlarging mandibular aneurysmal bone cyst with sclerotherapy and intralesional curettage. Cureus. 2025. https://doi.org/10.7759/cureus.78256.
8. Bakarman KA. Diagnosis and current treatment of aneurysmal bone cysts. Cureus. 2024;16:e53587. https://doi.org/10.7759/cureus.53587.
9. Zaghloul A, Haddad B, Khan W, Grimes L, Tucker K. A novel minimally invasive technique for treatment of unicameral bone cysts. Open Orthop J. 2015;9:475–9. https://doi.org/10.2174/1874325001509010475.
10. Samargandi R, Alkameshki M, Barnawi M, Alzahrani K, Iskander O, Nicolas Q, et al. Efficacy of percutaneous treatment of primary aneurysmal bone cysts (ABCs): A systematic review and meta-analysis. J Clin Med. 2023;12:7213. https://doi.org/10.3390/jcm12237213.
11. Varshney MK, Rastogi S, Khan SA, Trikha V. Is sclerotherapy better than intralesional excision for treating aneurysmal bone cysts? Clin Orthop Relat Res. 2010;468:1649–59.
12. Alhumaid I, Abu-Zaid A. Denosumab therapy in the management of aneurysmal bone cysts: a comprehensive literature review. Cureus. 2019;11:e3989. https://doi.org/10.7759/cureus.3989.
13. Kurucu N, Akyuz C, Ergen FB, Yalcin B, Kosemehmetoglu K, Ayvaz M, et al Denosumab treatment in aneurysmal bone cyst: Evaluation of nine cases. Pediatr Blood Cancer. 2018;65(4):e26926. https://doi.org/10.1002/pbc.26926.
14. Palmerini E, Ruggieri P, Angelini A, Boriani S, Campanacci D, Milano GM, et al. Denosumab in patients with aneurysmal bone cysts: A case series with preliminary results. Tumori. 2018;104:344–51. https://doi.org/10.1177/0300891618784808.
15. Maximen J, Robin F, Tronchot A, Rossetti A, Ropars M, Guggenbuhl P. Denosumab in the management of aneurysmal bone cyst. Jt Bone Spine. 2022;89:105260. https://doi.org/10.1016/j.jbspin.2021.105260.
16. Cottalorda J, Kohler R, Sales de Gauzy J, Chotel F, Mazda K, Lefort G, et al. Epidemiology of aneurysmal bone cyst in children: a multicenter study and literature review. J Pediatr Orthop B. 2004;13:389–94. https://doi.org/10.1097/01202412-200411000-00008.
17. Zehetgruber H, Bittner B, Gruber D, Krepler P, Trieb K, Kotz R, et al. Prevalence of aneurysmal and solitary bone cysts in young patients. Clin Orthop Relat Res. 2005;439:136–43. https://doi.org/10.1097/01.blo.0000173256.85016.c4.
18. Nasri E, Reith JD. Aneurysmal bone cyst: a review. J Pathol Transl Med. 2023;57:81–7. https://doi.org/10.4132/jptm.2023.02.23.
19. Evans J, Shamrock AG, Blake J. Unicameral bone cyst. Treasure Island (FL); 2025.
20. Flores I, Hamilton M, Zanchin-Baldissera E, Uchoa-Vasconcelos A, Chaves-Tarquinio S, Neutzling-Gomes A Simple and aneurysmal bone cyst: Aspects of jaw pseudocysts based on an experience of Brazilian pathology service during 53 years. Med Oral Patol Oral y Cir Bucal. 2016. https://doi.org/10.4317/medoral.21551.
21. Rehman R, Dekhou A, Osto M, Agemy J, Chaaban A, Yuhan B, et al Aneurysmal bone cysts of the craniofacial origin: a systematic review. OTO Open. 2021. https://doi.org/10.1177/2473974X211052950.
22. Zhang Y, Qiu Y, Zhang X, He X, Chen C, Chen M, et al. USP6-associated soft tissue tumors with bone metaplasia: Clinicopathologic and genetic analysis and the identification of novel USP6 fusion partners. Front Oncol. 2022;12:1065071. https://doi.org/10.3389/fonc.2022.1065071.
23. Oliveira AM, Perez-Atayde AR, Inwards CY, Medeiros F, Derr V, Hsi B-L, et al. USP6 and CDH11 Oncogenes identify the neoplastic cell in primary aneurysmal bone cysts and are absent in so-called secondary aneurysmal bone cysts. Am J Pathol. 2004;165:1773–80. https://doi.org/10.1016/S0002-9440(10)63432-3.
24. Althof PA, Ohmori K, Zhou M, Bailey JM, Bridge RS, Nelson M, et al. Cytogenetic and molecular cytogenetic findings in 43 aneurysmal bone cysts: aberrations of 17p mapped to 17p13.2 by fluorescence in situ hybridization. Mod Pathol Off J United States Can Acad Pathol Inc. 2004;17:518–25. https://doi.org/10.1038/modpathol.3800090.
25. Syed S, Painda MYK, Ghafoor D, Gu D, Wang F. Physiological roles and therapeutic implications of USP6. Cell Death Discov. 2025;11:231. https://doi.org/10.1038/s41420-025-02466-0.

26. Noordin S, Allana S, Umer M, Jamil M, Hilal K, Uddin N. Unicameral bone cysts: Current concepts. Ann Med Surg. 2018;34:43–9. https://doi.org/10.1016/j.amsu.2018.06.005.
27. Pižem J, Šekoranja D, Zupan A, Boštjančič E, Matjašič A, Mavčič B, et al. FUS-NFATC2 or EWSR1-NFATC2 fusions are present in a large proportion of simple bone cysts. Am J Surg Pathol. 2020;44:1623–34. https://doi.org/10.1097/PAS.0000000000001584.
28. Pižem J, Šekoranja D, Matjašič A, Zupan A, Boštjančič E, Limpel Novak KA, et al. The role of molecular diagnostics in aneurysmal and simple bone cysts - a prospective analysis of 19 lesions. Virchows Arch. 2021;479:795–802. https://doi.org/10.1007/s00428-021-03130-5.
29. Bhattacharyya I, Islam MN, Chehal HK, Reith JD, Islam S, Fernandes R, et al. Unusually large and aggressive traumatic bone cysts of the jaws – a series of five cases. Oral Surg. 2014;7:40–7. https://doi.org/10.1111/ors.12099.
30. Shindell R, Huurman WW, Lippiello L, Connolly JF. Prostaglandin levels in unicameral bone cysts treated by intralesional steroid injection. J Pediatr Orthop. 1989;9:516–9. https://doi.org/10.1097/01241398-198909010-00003.
31. Gerasimov AM, Toporova SM, Furtseva LN, Berezhnoy AP, Vilensky EV, Alekseeva RI. The role of lysosomes in the pathogenesis of unicameral bone cysts. Clin Orthop Relat Res. 1991;266:53–63.
32. Yu J, Chang S-S, Suratwala S, Chung W-S, Abdelmessieh P, Lee H-J, et al. Zoledronate induces apoptosis in cells from fibro-cellular membrane of unicameral bone cyst (UBC). J Orthop Res Off Publ Orthop Res Soc. 2005;23:1004–12. https://doi.org/10.1016/j.orthres.2005.02.012.
33. Lokiec F, Wientroub S. Simple bone cyst: Etiology, classification, pathology, and treatment modalities. J Pediatr Orthop B. 1998;7(4):262-273.
34. Usmanova M Differential x-ray diagnostics of aneurysmal and simple bone cyst. Iran J Radiol. 2014. https://doi.org/10.5812/iranjradiol.21310.
35. Park HY, Yang SK, Sheppard WL, Hegde V, Zoller SD, Nelson SD, et al. Current management of aneurysmal bone cysts. Curr Rev Musculoskelet Med. 2016;9:435–44. https://doi.org/10.1007/s12178-016-9371-6.
36. Rapp TB, Ward JP, Alaia MJ. Aneurysmal bone cyst. J Am Acad Orthop Surg. 2012;20:233–41. https://doi.org/10.5435/JAAOS-20-04-233.
37. Mahnken A, Nolte-Ernsting C, Wildberger J, Heussen N, Adam G, Wirtz D, et al. Aneurysmal bone cyst: value of MR imaging and conventional radiography. Eur Radiol. 2003;13:1118–24. https://doi.org/10.1007/s00330-002-1668-8.
38. Josip E, Kremser C, Haider B, Thaler M, Dammerer D, Henninger B. MRI analysis of simple and aneurysmal bone cysts in the proximal humerus: what actually matters in clinical routine. Skeletal Radiol. 2022;51:819–27. https://doi.org/10.1007/s00256-021-03889-3.
39. Olvi LG, Lembo GM, Velan O, Santini-Araujo E. Simple bone cyst. Tumors and tumor-like lesions of bone. London: Springer London; 2015. pp. 611–33. https://doi.org/10.1007/978-1-4471-6578-1_47.
40. Subramanian S, Kemp AK, Viswanathan VK. Bone cyst. Treasure Island (FL); 2025.
41. Zhang Y, Zhang C, Liu J, Han J. Aneurysmal bone cyst of the jaws: A retrospective analysis of 19 cases. Int Dent J. 2025;75:1988–98. https://doi.org/10.1016/j.identj.2025.01.006.
42. Palmisciano P, Hunter M, Lokesh N, Bin Alamer O, Scalia G, Giammalva GR, et al. Aneurysmal bone cyst of the spine in adult patients: A systematic review and comparison of primary vs secondary lesions. J Clin Neurosci. 2022;100:15–22. https://doi.org/10.1016/j.jocn.2022.03.040.
43. Wang Y, Tang F, Li Z, Chen Q. Pseudocysts of the jaw: a retrospective study of 41 cases from a single institution. BMC Oral Health. 2023;23:87. https://doi.org/10.1186/s12903-023-02741-5.
44. Vasconcelos A, de Bannak P, Naves IBPE, Azevedo IRGD, Tessarin RG incidence and major clinical approaches of aneurismatic bone cysts: a concise systematic review. MedNEXT J Med Heal Sci. 2022. https://doi.org/10.54448/mdnt22S603.
45. Das KN, Mehrotra A, Keshri A, Sinha M, Hameed N, Chidambaram K, et al. Long-term outcomes of temporal bone aneurysmal bone cysts: ambispective study with systematic review and pooled analysis. Otolaryngol Neck Surg. 2025;172:1493–501. https://doi.org/10.1002/ohn.1073.
46. Pala E, Trovarelli G, Angelini A, Cerchiaro MC, Ruggieri P. Modern treatment of unicameral and aneurysmatic bone cysts. EFORT open Rev. 2024;9:387–92. https://doi.org/10.1530/EOR-24-0027.
47. Döring K, Puchner S, Vertesich K, Funovics PT, Hobusch G, Sulzbacher I, et al. Results in the surgical treatment of aneurysmal bone cysts – A retrospective data analysis. Orthop Traumatol Surg Res. 2022;108:103095. https://doi.org/10.1016/j.otsr.2021.103095.
48. Deventer N, Schulze M, Gosheger G, de Vaal M, Deventer N. Primary aneurysmal bone cyst and its recent treatment options: a comparative review of 74 cases. Cancers (Basel). 2021;13:2362. https://doi.org/10.3390/cancers13102362.
49. Deventer N, Toporowski G, Gosheger G, de Vaal M, Luebben T, Budny T, et al. Aneurysmal bone cyst of the foot: A series of 10 cases. Foot Ankle Surg. 2022;28:276–80. https://doi.org/10.1016/j.fas.2021.03.002.
50. Yenigül AE, Sofulu Ö, Erol B Treatment of locally aggressive benign bone tumors by means of extended intralesional curettage without chemical adjuvants. SAGE Open Med. 2022. https://doi.org/10.1177/20503121221094199.
51. Strohm JA, Strohm PC, Kühle J, Schmal H, Zwingmann J. Management of juvenile and aneurysmal bone cysts: a systematic literature review with meta-analysis. Eur J Trauma Emerg Surg. 2023;49:361–72. https://doi.org/10.1007/s00068-022-02077-9.
52. Peeters SP, Van der Geest ICM, de Rooy JWJ, Veth RPH, Schreuder HWB. Aneurysmal bone cyst: The role of cryosurgery as local adjuvant treatment. J Surg Oncol. 2009;100:719–24. https://doi.org/10.1002/jso.21410.
53. Schreuder HW, Veth RP, Pruszczynski M, Lemmens JA, Koops HS, Molenaar WM. Aneurysmal bone cysts treated by curettage, cryotherapy and bone grafting. J Bone Joint Surg Br. 1997;79:20–5. https://doi.org/10.1302/0301-620x.79b1.7097.
54. Steffner RJ, Liao C, Stacy G, Atanda A, Attar S, Avedian R, et al. Factors associated with recurrence of primary aneurysmal bone cysts: Is argon beam coagulation an effective adjuvant treatment? J Bone Jt Surgery-American Vol. 2011;93:e1221–9. https://doi.org/10.2106/JBJS.J.01067.
55. Grahneis F, Klein A, Baur-Melnyk A, Knösel T, Birkenmaier C, Jansson V, et al. Aneurysmal bone cyst: A review of 65 patients. J Bone Oncol. 2019;18:100255. https://doi.org/10.1016/j.jbo.2019.100255.
56. Andreani L, Ipponi E, Serrano E, De Franco S, Cordoni M, Bechini E, et al. Aneurysmal bone cyst of the pelvis in children and adolescents: Effectiveness of surgical treatment with curettage, cryotherapy and bone grafting. Healthcare. 2023;11:2658. https://doi.org/10.3390/healthcare11192658.
57. Flont P, Kolacinska-Flont M, Niedzielski K. A comparison of cyst wall curettage and en bloc excision in the treatment of aneurysmal bone cysts. World J Surg Oncol. 2013;11:109. https://doi.org/10.1186/1477-7819-11-109.
58. Deventer N, Deventer N, Gosheger G, de Vaal M, Budny T, Luebben T, et al. Evaluation of different treatment modalities for fractured and non-fractured simple bone cyst. Med (Baltim). 2021;100:e26703. https://doi.org/10.1097/MD.0000000000026703.
59. Hajialiloo Sami S, Toloue Ghamari B, Kargar Shooroki K, Mohammadi Aniloo F, Ammar W, Rikhtehgar M, et al. Metatarsal aneurysmal bone cysts treated with en bloc resection and reconstruction with fibular allograft. Foot Ankle Int. 2025;46:9–16. https://doi.org/10.1177/10711007241287714.
60. Jamshidi K, Haji Agha Bozorgi M, Hajializade M, Bagherifard A, Mirzaei A. Tailored treatment of aneurysmal bone cyst of the scapula: en bloc resection for the body and extended curettage for the neck and acromion. J Shoulder Elb Surg. 2020;29:961–7. https://doi.org/10.1016/j.jse.2019.09.015.
61. Zhao Y, He S, Sun H, Cai X, Gao X, Wang P, et al. Symptomatic aneurysmal bone cysts of the spine: clinical features, surgical outcomes, and prognostic factors. Eur Spine J. 2019;28:1537–45. https://doi.org/10.1007/s00586-019-05920-7.
62. Costa D, Di, Gabrielli E, Cerrone M, Di Gialleonardo E, Maccauro G, Vitiello R. Pathological fractures in aneurysmal bone cysts: A systematic review. J Clin Med. 2024;13:2485. https://doi.org/10.3390/jcm13092485.
63. Kececi B. Effect of adjuvant therapies on recurrence in aneurysmal bone cysts. ACTA Orthop Traumatol Turc. 2014;48:500–6. https://doi.org/10.3944/AOTT.2014.14.0020.
64. Myilvahanan G, Shenoy S, Thamba D, Kumaravel A. Extended intralesional curettage with bone grafting for primary aneurysmal bone cyst of the talus -A report of two cases with long-term follow-up. J Orthop Rep. 2024;3:100291. https://doi.org/10.1016/j.jorep.2023.100291.
65. Crowe MM, Houdek MT, Moran SL, Kakar S. Aneurysmal bone cysts of the hand, wrist, and forearm. J Hand Surg Am. 2015;40:2052–7. https://doi.org/10.1016/j.jhsa.2015.06.108.
66. Wang EHM, Marfori ML, Serrano MVT, Rubio DA. Is curettage and high-speed burring sufficient treatment for aneurysmal bone cysts? Clin Orthop Relat Res. 2014;472:3483–8. https://doi.org/10.1007/s11999-014-3809-1.
67. van Geloven TPG, van de Sande MAJ, van der Heijden L. The treatment of aneurysmal bone cysts. Curr Opin Pediatr. 2023;35:131–7. https://doi.org/10.1097/MOP.0000000000001205.
68. Cummings JE, Smith RA, Heck RK. Argon beam coagulation as adjuvant treatment after curettage of aneurysmal bone cysts: A preliminary study. Clin

Orthop Relat Res. 2010;468:231–7. https://doi.org/10.1007/s11999-009-0914-7.
69. Rahul P, Jaymin S, A SA, Ashokkumar S, Jyotindra P, Shashank P. Percutaneous sclerotherapy for the treatment of aneurysmal bone cyst, what are the outcomes ? Gujarat Cancer Soc Res J. 2018;20:54–7.
70. Oliveira MBDR, Meohas W, Silva RR, de Carvalho GS, Mello FC, de Paschoal Q, Percutaneous treatment of aneurysmal bone cyst with calcitonin and methylprednisolone. Acta Ortop Bras. 2018;26:314–9. https://doi.org/10.1590/1413-785220182605201423.
71. Chang CY, Kattapuram SV, Huang AJ, Simeone FJ, Torriani M, Bredella MA. Treatment of aneurysmal bone cysts by percutaneous CT-guided injection of calcitonin and steroid. Skeletal Radiol. 2017;46:35–40. https://doi.org/10.1007/s00256-016-2503-7.
72. Rathod DJ, Prajapati DD, Vora DD. A study of 16 patients with aneurysmal bone cyst treated with intralesional methyl prednisolone. Int J Orthop Sci. 2022;8:07–9. https://doi.org/10.22271/ortho.2022.v8.i2a.3108.
73. Abdel-Ghany I Bone marrow injection for treatment of aneurysmal bone cyst. MOJ Orthop Rheumatol. 2016. . https://doi.org/10.15406/mojor.2016.05.00182.
74. Exner GU, Schai PA. Treatment of unicameral and aneurysmal bone cysts by minimally invasive percutaneous injection of grafton DBF putty using the kyphon cement delivery system. Open J Orthop. 2020;10:143–51. https://doi.org/10.4236/ojo.2020.107019.
75. Cevolani L, Campanacci L, Sambri A, Lucarelli E, De Paolis M, Donati DM. Is percutaneous injection of bone marrow concentrate, demineralized bone matrix and PRF an alternative to curettage and bone grafting for treating aneurysmal bone cyst? J Tissue Eng Regen Med. 2021;15:269–78. https://doi.org/10.1002/term.3175.
76. Meirlaen S, Haoudou R, Thiteux Q, Bellanova L, Docquier P-L. Treatment of bone cysts by percutaneous injection of demineralized bone matrix mixed with bone marrow. Acta Orthop Belg. 2022;88:559–67. https://doi.org/10.52628/88.3.10051.
77. Fernandez de Grado G, Keller L, Idoux-Gillet Y, Wagner Q, Musset A-M, Benkirane-Jessel N, et al. Bone substitutes: a review of their characteristics, clinical use, and perspectives for large bone defects management. J Tissue Eng. 2018;9:2041731418776819. https://doi.org/10.1177/2041731418776819.
78. Battistelli M, Rapisarda A, Montano N, Pedicelli A, Valente I, Olivi A, et al. Successful treatment of a C2 aneurysmal bone cyst with hydroxyapatite and calcium sulfate synthetic bone void filler injection: a case report. J Spine Surg. 2024;10:745–57. https://doi.org/10.21037/jss-24-16.
79. Brosjö O, Pechon P, Hesla A, Tsagozis P, Bauer H. Sclerotherapy with polidocanol for treatment of aneurysmal bone cysts. Acta Orthop. 2013;84:502–5. https://doi.org/10.3109/17453674.2013.850013.
80. Shiels WE, Mayerson JL. Percutaneous doxycycline treatment of aneurysmal bone cysts with low recurrence rate: A preliminary report. Clin Orthop Relat Res. 2013;471:2675–83. https://doi.org/10.1007/s11999-013-3043-2.
81. Kumar D, Kumar S, Kumar D, Patel BM, Kumar A, Kumar S, et al. Sclerotherapy for aneurysmal bone cyst: A single-center experience. Cureus. 2021;13:e18469. https://doi.org/10.7759/cureus.18469.
82. Weber KS, Jensen CL, Petersen MM. Sclerotherapy as a primary or salvage procedure for aneurysmal bone cysts: A single-center experience. World J Orthop. 2023;14:698–706. https://doi.org/10.5312/wjo.v14.i9.698.
83. Ulici A, Florea D-C, Carp M, Ladaru A, Tevanov I. Treatment of the aneurysmal bone cyst by percutaneous intracystic sclerotherapy using ethanol 95% in children. Int Orthop. 2018;42:1413–9. https://doi.org/10.1007/s00264-018-3841-y.
84. De Cristofaro R, Biagini R, Boriani S, Ricci S, Ruggieri P, Rossi G, et al. Selective arterial embolization in the treatment of aneurysmal bone cyst and angioma of bone. Skeletal Radiol. 1992;21:523–7. https://doi.org/10.1007/BF00195235.
85. Celebioglu E, Karaca M, Basarir K, Yildiz H, Bilgic S. Selective arterial embolization (SAE) in aneurysmal bone cyst (ABC). Ann Med Res. 2021;28:743. https://doi.org/10.5455/annalsmedres.2020.11.1141.
86. Rossi G, Rimondi E, Bartalena T, Gerardi A, Alberghini M, Staals EL, et al. Selective arterial embolization of 36 aneurysmal bone cysts of the skeleton with N-2-butyl cyanoacrylate. Skeletal Radiol. 2010;39:161–7. https://doi.org/10.1007/s00256-009-0757-z.
87. Zarzour OA, Santiago FR, Serrano NO, Abdallah AH, El-Sharkawy MA, Mourad AF. CT-guided radiofrequency ablation in patients with aneurysmal bone cysts. Eur J Radiol. 2018;100:116–23. https://doi.org/10.1016/j.ejrad.2018.01.020.
88. Tsoumakidou G, Koch G, Caudrelier J, Garnon J, Cazzato RL, Edalat F, et al. Image-guided spinal ablation: A review. Cardiovasc Intervent Radiol. 2016;39:1229–38. https://doi.org/10.1007/s00270-016-1402-6.
89. Zellner M, Kellenberger CJ, Pistorius S, Dreher T, Pfammatter T, Knüsel P, et al. Technical feasibility and outcome of cryoablation of aneurysmal bone cysts in pediatric patients. J Vasc Interv Radiol. 2024;35:1655–e16612. https://doi.org/10.1016/j.jvir.2024.07.027.
90. Fritz J, Sonnow L, Morris CD. Adjuvant MRI-guided percutaneous cryoablation treatment for aneurysmal bone cyst. Skeletal Radiol. 2019;48:1149–53. https://doi.org/10.1007/s00256-018-3115-1.
91. Aiba H, Kobayashi M, Waguri-Nagaya Y, Goto H, Mizutani J, Yamada S, et al. Treatment of aneurysmal bone cysts using endoscopic curettage. BMC Musculoskelet Disord. 2018;19:268. https://doi.org/10.1186/s12891-018-2176-6.
92. Megremis P, Megremis O. Endoscopic high-speed curettage and bone grafting of aneurysmal bone cyst of the lateral malleolus in a 9-year-old boy. J Orthop Rep. 2022;1:100091. https://doi.org/10.1016/j.jorep.2022.100091.
93. Cruz GS, Cuevas-Suárez CE, Saavedra JPA, Giorgis R, Teixeira MRK, Muniz FWMG. Percutaneous treatments of primary aneurysmal bone cysts: systematic review and meta-analysis. Eur J Orthop Surg Traumatol. 2021;31:1287–95. https://doi.org/10.1007/s00590-021-02893-6.
94. Parker J, Soltani S, Boissiere L, Obeid I, Gille O, Kieser DC. Spinal aneurysmal bone cysts (ABCs): Optimal management. Orthop Res Rev. 2019;11:159–66. https://doi.org/10.2147/ORR.S211834.
95. Gettleman BS, Padilla AN, Kumar S, Wren TAL, Miller J, Pawel BR, et al. Use of surgical adjuvants does not decrease recurrence of aneurysmal bone cysts in surgical intervention with pediatric patients. J Pediatr Orthop. 2024;44:e79–83. https://doi.org/10.1097/BPO.0000000000002536.
96. Levanon E, Merose O, Segal O, Dadia S, Sternheim A, Levin D, et al. Does cryotherapy decrease the local recurrence rate in the treatment of an aneurysmal bone cyst? A comparative assessment. J Orthop Res. 2024;42:1369–75. https://doi.org/10.1002/jor.25775.
97. Erol B, Topkar MO, Caliskan E, Erbolukbas R. Surgical treatment of active or aggressive aneurysmal bone cysts in children. J Pediatr Orthop B. 2015;24:461–8. https://doi.org/10.1097/BPB.0000000000000173.
98. Puthoor D, Francis L, Ismail R. Is sclerotherapy with polidocanol a better treatment option for aneurysmal bone cyst compared to conventional curettage and bone grafting? J Orthop. 2021;25:265–70. https://doi.org/10.1016/j.jor.2021.05.020.
99. Dormans JP, Hanna BG, Johnston DR, Khurana JS. Surgical treatment and recurrence rate of aneurysmal bone cysts in children. Clin Orthop Relat Res. 2004;421:205–11. https://doi.org/10.1097/01.blo.0000126336.46604.e1.
100. Yang J, Zhang J, Guo Y. Aneurysmal bone cyst: comparison of percutaneous steroid injections and curettage. Chin J Pediatr Surg 34: 353–6. https://doi.org/10.3760/cma.j.issn.0253-3006.2013.05.009
101. Masthoff M, Gerwing M, Schneider KN, Köhler M, Deventer N, Schindler P, et al. Combined transarterial embolization and percutaneous sclerotherapy as treatment for refractory and nonresectable aneurysmal bone cysts. J Vasc Interv Radiol. 2021;32:1425–e14342. https://doi.org/10.1016/j.jvir.2021.07.008.
102. Green NM, Pagkalos J, Jeys LM, Stevenson JD, Gregory JJ. Humeral Simple Bone Cysts: Observational Versus Interventional Management. J Pediatr Orthop. 2019;39:e472–7. https://doi.org/10.1097/BPO.0000000000001344.
103. Da Costa Mendes TC, De Souza GC, De Lima Pinheiro BC, De Araujo Vasconc AJ Elos, Cabral LN, Conservative treatment of simple bone cyst in the mandible, et al. Oral Surg Oral Med Oral Pathol Oral Radiol. 2020;129:e61. https://doi.org/10.1016/j.oooo.2019.06.229.
104. Mylle J, Burssens A, Fabry G. Simple bone cysts. Arch Orthop Trauma Surg. 1992;111:297–300. https://doi.org/10.1007/BF00420053.
105. Griauzde J, Gemmete JJ, Farley F. Successful treatment of a musculoskeletal tumor society grade 3 aneurysmal bone cyst with N-butyl cyanoacrylate embolization and percutaneous cryoablation. J Vasc Interv Radiol. 2015;26:905–9. https://doi.org/10.1016/j.jvir.2015.01.019.
106. Machak GN, Snetkov AI. The impact of curettage technique on local control in giant cell tumour of bone. Int Orthop. 2021;45:779–89. https://doi.org/10.1007/s00264-020-04860-y.
107. Smolle MA, Roessl V, Leithner A. Effect of local adjuvants following curettage of benign and intermediate tumours of bone: A systematic review of the literature. Cancers (Basel). 2023;15:4258. https://doi.org/10.3390/cancers15174258.
108. Zuo D, Zheng L, Sun W, Fu D, Hua Y, Cai Z. Contemporary adjuvant polymethyl methacrylate cementation optimally limits recurrence in primary giant cell tumor of bone patients compared to bone grafting: a systematic review and meta-analysis. World J Surg Oncol. 2013;11:156. https://doi.org/10.1186/1477-7819-11-156.

109. Costello JP, Travis LM, Jahn J, Pretell-Mazzini JA The role of bone grafting vs. Bone cement in the treatment of giant cell tumor of bone. JBJS Rev. 2024. . https://doi.org/10.2106/JBJS.RVW.24.00080.
110. Alshryda S, Wright J. Evidence-based treatment of simple bone cyst. Paediatric orthopaedics. Cham: Springer International Publishing; 2017. pp. 419–32. https://doi.org/10.1007/978-3-319-41142-2_43.
111. Masquijo JJ, Baroni E, Miscione H. Continuous decompression with intramedullary nailing for the treatment of unicameral bone cysts. J Child Orthop. 2008;2:279–83. https://doi.org/10.1007/s11832-008-0114-0.
112. Traub F, Eberhardt O, Fernandez FF, Wirth T. Solitary bone cyst: a comparison of treatment options with special reference to their long-term outcome. BMC Musculoskelet Disord. 2016;17:162. https://doi.org/10.1186/s12891-016-1012-0.
113. Vasilescu DE, Cosma D. Elastic stable intramedullary nailing for fractures in children - principles, indications, surgical technique. Clujul Med. 2014;87:91–4. https://doi.org/10.15386/cjmed-274.
114. de Sanctis N, Andreacchio A. Elastic stable intramedullary nailing is the best treatment of unicameral bone cysts of the long bones in children? J Pediatr Orthop. 2006;26:520–5. https://doi.org/10.1097/01.bpo.0000217729.39288.df.
115. Pogorelić Z, Furlan D, Biočić M, Meštrović J, Jurić I, Todorić D. Titanium intramedullary nailing for treatment of simple bone cysts of the long bones in children. Scott Med J. 2010;55:35–8. https://doi.org/10.1258/rsmsmj.55.3.35.
116. Zhang P, Zhu N, Du L, Zheng J, Hu S, Xu B. Treatment of simple bone cysts of the humerus by intramedullary nailing and steroid injection. BMC Musculoskelet Disord. 2020;21:70. https://doi.org/10.1186/s12891-020-3054-6.
117. Wang X, Han J, Li Y, Liu Y, Luo J. Comparative efficacy and safety profile for the treatment of humeral bone cysts in children: curettage and mixed bone grafting either with or without elastic intramedullary nailing. J Orthop Surg Res. 2021;16:241. https://doi.org/10.1186/s13018-020-02130-6.
118. Zhang P, Zhu N, Du L, Zheng J, Hu S, Xu B. Treatment of simple bone cysts of humerus by intramedullary decompression and drainage with elastic stable intramedullary nailing combined with intralesional injections of steroids. 2019. https://doi.org/10.21203/rs.2.11547/v1
119. Kanellopoulos AD, Mavrogenis AF, Papagelopoulos PJ, Soucacos PN. Elastic intramedullary nailing and DBM-Bone marrow injection for the treatment of simple bone cysts. World J Surg Oncol. 2007;5:111. https://doi.org/10.1186/1477-7819-5-111.
120. Li J, Rai S, Ze R, Tang X, Liu R, Hong P. Injectable calcium sulfate vs mixed bone graft of autologous iliac bone and allogeneic bone. Med (Baltim). 2020;99:e20563. https://doi.org/10.1097/MD.0000000000020563.
121. Lee TG, Ahmad TS. Intralesional autologous blood injection compared to corticosteroid injection for treatment of chronic plantar fasciitis. A prospective, randomized, controlled trial. Foot Ankle Int. 2007;28:984–90. https://doi.org/10.3113/FAI.2007.0984.
122. Shubham Shankar U, Khaitan R A. comparative study between single-dose intralesional autologous blood, and corticosteroid injection in chronic plantar fasciitis. – A short-term follow-up study. Int J Sci Res. 2024;20–1. https://doi.org/10.36106/ijsr/7606246.
123. Maximen J, Jeantet R-E, Violas P. Surgical management of proximal femoral unicameral bone cyst in children. Bone Joint J. 2024;106–B:508–14. https://doi.org/10.1302/0301-620X.106B5.BJJ-2023-0577.R4.
124. El Masry AM, Elkhateeb TM, Azmy SI, Abdelkader SG. Comparing indirect decompression using elastic nails versus curettage, bone graft substitute, and intramedullary stabilization for the management of unicameral bone cysts in skeletally immature patients: a retrospective case series study. Curr Orthop Pract. 2021;32:187–96. https://doi.org/10.1097/BCO.0000000000000965.
125. Li W, Xu R, Du M, Chen H. Comparison of titanium elastic intramedullary nailing versus injection of bone marrow in treatment of simple bone cysts in children: a retrospective study. BMC Musculoskelet Disord. 2016;17:343. https://doi.org/10.1186/s12891-016-1184-7.
126. Bishay SNG. Curettage without bone grafting for a simple bone cyst in the capitate. J Orthop Surg. 2015;23:361–4. https://doi.org/10.1177/230949901502300322.
127. Canavese F, Wright JG, Cole WG, Hopyan S. Unicameral bone cysts. J Pediatr Orthop. 2011;31:50–5. https://doi.org/10.1097/BPO.0b013e3181ff7510.
128. Hashemi-Nejad A, Cole WG, Incomplete healing of simple bone cysts after, steroid injections. J Bone Joint Surg Br. 1997;79–B:727–30. https://doi.org/10.1302/0301-620X.79B5.0790727.
129. Wright JG, Yandow S, Donaldson S, Marley L. A randomized clinical trial comparing intralesional bone marrow and steroid injections for simple bone cysts. J Bone Jt Surgery-American Vol. 2008;90:722–30. https://doi.org/10.2106/JBJS.G.00620.
130. D'Amato RD, Memeo A, Fusini F, Panuccio E, Peretti G. Treatment of simple bone cyst with bone marrow concentrate and equine-derived demineralized bone matrix injection versus methylprednisolone acetate injections: A retrospective comparative study. Acta Orthop Traumatol Turc. 2020;54:49–58. https://doi.org/10.5152/j.aott.2020.01.371.
131. Mohamed MAHD, Mostafa MAR, Elseddawy AM role of denosumab in the management of aneurysmal bone cysts. QJM Int J Med. 2023. . https://doi.org/10.1093/qjmed/hcad069.613.
132. Narayanan P, Denosumab. A comprehensive review. South Asian J Cancer. 2013;2:272–7. https://doi.org/10.4103/2278-330X.119895.
133. Ono T, Hayashi M, Sasaki F, Nakashima T. RANKL biology: bone metabolism, the immune system, and beyond. Inflamm Regen. 2020;40:2. https://doi.org/10.1186/s41232-019-0111-3.
134. Evangelisti G, Altorfer FCS, Falzetti L, Palmerini E, Griffoni C, Ghermandi R, et al. Denosumab re-challenge and long-term efficacy for aneurysmal bone cyst of the spine: Enhanced treatment algorithm. J Clin Med. 2024;13:4522. https://doi.org/10.3390/jcm13154522.
135. Kotaka S, Fujiwara Y, Ota R, Manabe H, Adachi N. The efficacy of denosumab in treating spinal aneurysmal bone cyst: A case report. Cureus. 2023. https://doi.org/10.7759/cureus.39954.
136. El Masry AM, Azmy SI, Mustafa MAR, Abuelhadid MA. Using denosumab as a nonsurgical management of aneurysmal bone cysts in the pelvis. Clin Orthop Surg. 2024;16:149–56. https://doi.org/10.4055/cios22228.
137. Hung YP, Bredella MA, Lobmaier IVK, Lozano-Calderón SA, Rosenberg AE, Nielsen GP. Aneurysmal bone cyst and osteoblastoma after neoadjuvant denosumab: histologic spectrum and potential diagnostic pitfalls. APMIS. 2022;130:206–14. https://doi.org/10.1111/apm.13211.
138. Celayir A, Sahin E, Ozsahin MK, Botanlioglu H. Postoperative denosumab therapy in aneurysmal bone cysts and osteoblastomas: A case series and review of current literature. Cureus. 2025;17:e83997. https://doi.org/10.7759/cureus.83997.
139. Vanderniet JA, Tsinas D, Wall C-L, Girgis CM, London K, Keane C, et al. Surgical management and denosumab for aneurysmal bone cysts of the spine in an australian tertiary paediatric centre. Calcif Tissue Int. 2023;112:592–602. https://doi.org/10.1007/s00223-023-01068-1.
140. Del Sindaco G, Berlanga P, Brugières L, Thebault E, Mantovani G, Wicart P, et al Mineral and bone consequences of high dose denosumab therapy to treat an aneurysmal bone cyst, a child case report. Front Endocrinol (Lausanne). 2021. https://doi.org/10.3389/fendo.2021.698963.
141. Sandhu V, Ajit Singh V, Puri A Exploring denosumab's potential in aneurysmal bone cyst treatment: A scoping review. J Orthop Surg. 2024. https://doi.org/10.1177/10225536241297105.
142. Dürr HR, Grahneis F, Baur-Melnyk A, Knösel T, Birkenmaier C, Jansson V, et al. Aneurysmal bone cyst: results of an off label treatment with Denosumab. BMC Musculoskelet Disord. 2019;20:456. https://doi.org/10.1186/s12891-019-2855-y.
143. Skubitz KM, Peltola JC, Santos ER, Cheng EY. Response of aneurysmal bone cyst to denosumab. Spine (Phila Pa 1976). 2015;40:E1201–4. https://doi.org/10.1097/BRS.0000000000001027.
144. Higuchi T, Yamamoto N, Shirai T, Hayashi K, Takeuchi A, Kimura H, et al. Treatment outcomes of the simple bone cyst. Med (Baltim). 2018;97:e0572. https://doi.org/10.1097/MD.0000000000010572.
145. Noordin S, Ahmad T, Umer M, Allana S, Hilal K, Uddin N, et al. Aneurysmal bone cyst of the pelvis and extremities: Contemporary management. Int J Surg Oncol. 2019;4:e71–71. https://doi.org/10.1097/IJ9.0000000000000071.
146. Muratori F, Scanferla R, Roselli G, Frenos F, Campanacci DA. Long term outcome of surgical treatment of chondroblastoma: analysis of local control and growth plate/articular cartilage related complications. BMC Musculoskelet Disord. 2023;24:139. https://doi.org/10.1186/s12891-023-06239-7.
147. Kumar BY, Thirumal R, Chander SG. Aneurysmal bone cyst of thoracic spine with neurological deficit and its recurrence treated with multimodal intervention – A case report. Surg Neurol Int. 2020;11:274. https://doi.org/10.25259/SNI_466_2020.
148. Milton R, Kalanjiyam GP, Shetty SR, Kanna AP. Dural injury following elective spine surgery – A prospective analysis of risk factors, management and complications. J Clin Orthop Trauma. 2023;41:102172. https://doi.org/10.1016/j.jcot.2023.102172.
149. Alqahtani N, Altwalah J, Alkhalifah A, Garad F, Alahmari F, Alrashidi I. Selective arterial embolization of aneurysmal bone cyst in the pubic bone: A possible

primary treatment. Radiol Case Rep. 2021;16:1280–3. https://doi.org/10.1016/j.radcr.2021.02.054.
150. Lin PP, Brown C, Raymond KA, Deavers MT, Yasko AW. Aneurysmal bone cysts recur at juxtaphyseal locations in skeletally immature patients. Clin Orthop Relat Res. 2008;466:722–8. https://doi.org/10.1007/s11999-007-0080-8.
151. UÇAN V, PULATKAN A, ERDEM AC. Results of the local steroid injections in the simple bone cyst. Bezmialem Sci. 2020;8:68–72. https://doi.org/10.14235/bas.galenos.2019.3121.
152. Cevolani L, Staals E, Campanacci L, Dozza B, Martella C, Spinnato P, et al. Aneurysmal bone cyst: Is selective arterial embolization effective as curettage and bone grafting? J Surg Oncol. 2023;128:1428–36. https://doi.org/10.1002/jso.27422.
153. Balko J, Golas W, Kaspar L, Krskova L, Strnadova M, Kotis J, et al. Novel and unusual USP6 fusion partners in aneurysmal bone cyst and their role in pathogenesis and histopathological evaluation of this disease. J Clin Pathol. 2025;78:399–403. https://doi.org/10.1136/jcp-2023-209306.
154. Cottalorda J, Louahem Sabah D, Joly Monrigal P, Jeandel C, Delpont M. Minimally invasive treatment of aneurysmal bone cysts: Systematic literature review. Orthop Traumatol Surg Res. 2022;108:103272. https://doi.org/10.1016/j.otsr.2022.103272.
155. Horiuchi K, Kobayashi E, Mizuno T, Susa M, Chiba K. Hypercalcemia following discontinuation of denosumab therapy: A systematic review. Bone Rep. 2021;15:101148. https://doi.org/10.1016/j.bonr.2021.101148.
156. Donaldson, S., Chundamala, J., Yandow, S., & Wright, J. G. Treatment for unicameral bone cysts in long bones: an evidence based review. Orthop Rev. 2010. https://doi.org/10.4081/or.2010.e13.

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